

In addition, REITs are levered while NCREIF returns are reported on an unlevered basis. Researchers have developed many methods to move from appraisals to transaction-level (and repeat-sales) indices, to desmooth real estate returns, to take into account selection bias, and to remove other illiquidity biases. The effects of these biases are enormous. Lin and Vandell (2012), for example, report that for a one-year holding period, the variance of direct real estate returns should be three times higher than the raw variance reported in Table 11.11. Pagliari, Scherer, and Monopoli (2005) argue that there is no difference in direct and indirect real estate returns after adjusting for leverage and sector composition and also adjusting direct real estate returns for appraisal smoothing.

Smoothing and selection biases are shared by all illiquid assets—including private equity—and I discuss methods of dealing with them in chapter 13. I use NCREIF returns for now, but you’ve been warned: these are not actual returns.

*In the Long Run, Direct and Indirect Real Estate Returns Move Together*

Direct and indirect real estate returns are linked in the long run. Figure 11.12 graphs the correlation of long-horizon log NCREIF returns with long-horizon log REIT returns. At the one-quarter horizon, the correlation is around 0.22, which reflects the low correlation between NCREIF and REIT returns in Table 11.11. (Quarterly returns in Table 11.11 are arithmetic returns but I use log returns in Figure 11.12; see the appendix.) After twelve quarters, the correlation tapers to

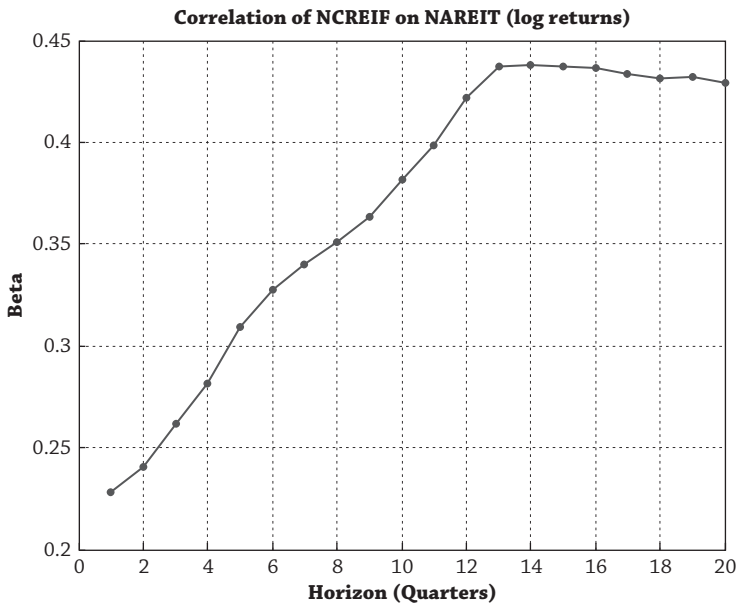


Figure 11.12